laugh at us saying we can't launch even a sounding rocket. Going ahead with the launch would mean that we could send the rocket up, but couldn't achieve the goal of igniting sodium vapour. There were four rockets waiting to be fired. And all were destined to fail. Murthy decided to go ahead with the launch.

'We can't afford to be laughed at. Let's go ahead,' he said. Later we would have to do a failure analysis to convince the world, though we knew before the failure the reason for it. We launched the four rockets across sixteen hours. They all failed, one after another. While everyone else remained tensed before each of the four launches, Murthy, Kalam and I waited for the inevitable—sad, composed and determined. Much later we successfully launched the Centaure rockets and did sodium vapour experiments by replacing G20 gunpowder with another explosive that did its job irrespective of the atmospheric pressure.

We were learning from every mistake so as not to repeat failures, but there was no dearth of new mistakes. And every time we fell, we made sure that we laughed at ourselves, dusted our backs and walked on to master another lesson.

While the Judi-Dart adventure cost us four rockets, some, like Experiment 45-01 were saved just in time. The year was 1967. The French Centaure sounding rocket had as its payload a transponder developed by O P N Calla and an electron probe by Satyaprakash. For the two things to be released from the rocket, the nose cone had to be separated at a predetermined altitude, and this depended on an ejection mechanism developed by A P J Abdul Kalam and me.

The idea was rudimentary: at the time of lift-off, a timer device would be set off. The timer would be set in such a way that when the rocket reaches an altitude of 63 km, it would activate the nose cone ejection mechanism. A squib would fire and separate the nose cone, letting the electron probe out. The leftovers of the rocket, with the probe and transponder would then plunge into the Arabian Sea. In the less than ten minutes of flight and fall, the probe would study electron density in the